EXPERIMENT 3

Design of Test Cases and Maintenance Plan for a

Smart Parking Management System

Aim

To design suitable test cases and prepare a maintenance plan for a Smart Parking Management System to ensure that the application works correctly and remains reliable after deployment.

Objective

- To understand the importance of software testing.
- To prepare test cases for the major functions of the application.
- To identify possible errors and verify the expected results.
- To prepare a maintenance plan for keeping the software efficient and secure.

Introduction

The Smart Parking Management System is an application that helps users find and reserve available parking spaces. Users can log in, check parking availability, book a slot, make payments, and receive booking notifications. The administrator can manage parking spaces and monitor parking activities.

Testing is an important part of developing this system because even a small error in slot availability or payment processing can cause problems for users. Maintenance is also required after deployment to fix bugs, improve performance, and add new features.

Modules Considered for Testing

The main modules considered for testing are:

1. User Registration and Login
2. Parking Slot Search
3. Parking Slot Reservation
4. Online Payment
5. Notification System
6. Admin Management

Test Cases

Test Case ID	Test Scenario	Input	Expected Result
TC01	User Registration	Valid user details	New account should be created
TC02	User Login	Correct username and password	User should be logged in successfully
TC03	Invalid Login	Wrong password	Error message should be displayed
TC04	Search Parking	Select parking location	Available slots should be displayed
TC05	Reserve Slot	Select an available slot	Slot should be reserved successfully
TC06	Book Occupied Slot	Select an occupied slot	Booking should not be allowed
TC07	Online Payment	Valid payment details	Payment should be completed successfully
TC08	Cancel Booking	Select an existing booking	Booking should be cancelled
TC09	Notification	Successful booking	Confirmation notification should be sent
TC10	Admin Report	Select report option	Parking report should be generated

Types of Testing

1. Unit Testing

Each individual module is tested separately. For example, the login module can be tested independently to check whether it accepts valid credentials and rejects incorrect ones.

2. Integration Testing

This checks whether different modules work properly together. For example, after a user books a parking slot, the payment module and booking module should update correctly.

3. System Testing

The complete application is tested as a single system to verify whether all its functions work according to the requirements.

4. User Acceptance Testing

The application is tested from the user's point of view to make sure that it is convenient and meets the actual parking requirements.

Maintenance Plan

Software maintenance is performed after the application is deployed. It helps keep the system functional, secure, and updated.

Maintenance Type	Purpose	Example
Corrective Maintenance	Fix existing errors	Fix incorrect parking availability
Adaptive Maintenance	Adapt to new environments	Support a new mobile OS version
Perfective Maintenance	Improve existing features	Improve the booking interface
Preventive Maintenance	Avoid future problems	Database backup and security updates

Maintenance Activities

- Fix software bugs reported by users.
- Regularly back up the parking database.
- Update security features to protect user information.
- Monitor application performance.
- Improve the user interface based on feedback.
- Add new features when required.
- Regularly check payment and notification services.

Maintenance Schedule

Activity	Frequency
Database Backup	Daily
Bug Fixing	Whenever Required
Security Updates	Monthly
Performance Monitoring	Monthly
Feature Improvements	Based on User Requirements

Result

The required test cases were designed for the major functions of the Smart Parking Management System. The test cases help verify important operations such as login, parking search, slot reservation, payment, and notifications. A maintenance plan was also prepared to ensure that the application continues to work properly after deployment.

Conclusion

Testing helps identify errors and confirms that the Smart Parking Management System works as expected. A proper maintenance plan helps keep the application secure, reliable, and easy to improve. Regular testing, bug fixing, backups, and updates will ensure better performance and a smoother experience for users.